SFU NatLangLab

CMPT 413/713: Natural Language Processing

# Conclusion

Fall 2022

2022-12-05

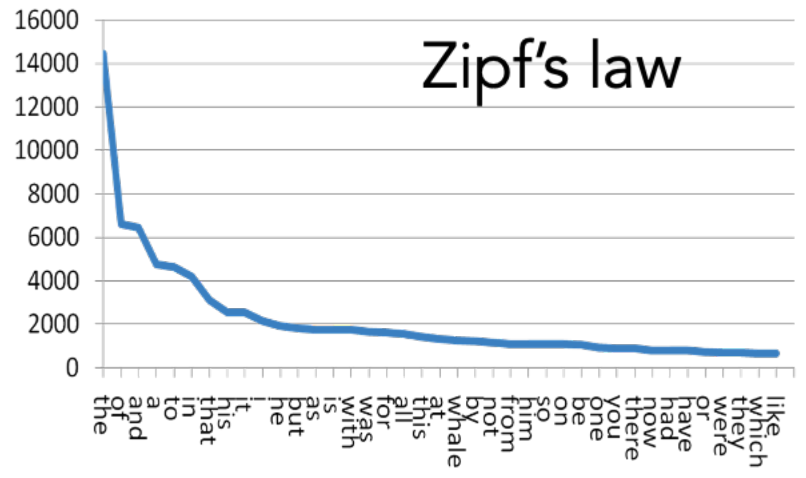
# What is natural language?

- Way for humans to communicate with each other
  - Use it to express feelings, transmit information, and store knowledge
- Discrete, combinations of symbols
- Learnable by all humans

# Key challenges of understanding language

I saw her duck

- Ambiguity (at multiple levels)
- Scale
- Sparsity (Long tail)
- Variation
- Expressivity
- Unmodeled Variables (Implicit knowledge / Context)
- Unknown representation (representation)



# What has this course covered?

**Representations**

- Embeddings
- Compositional
- Structured

**Methods**

- Rule based
- Statistical Methods
- Neural Models
- Dynamic programming
- Language Models
- Memory Networks

**Applications and Tasks**

- Machine Translation
- Question Answering
- Text Generation
- Dialogue Agents
- Grounding

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# Key Questions

- How to **represent** language?
- How to **compose** meaning?
- How do we **parse** the “surface form” of language to a representation that a computer can process?
- How do we take an encoded representation and **generate** text from it?
- What are the key methods / algorithms for **learning** these representations?

# Methods we covered

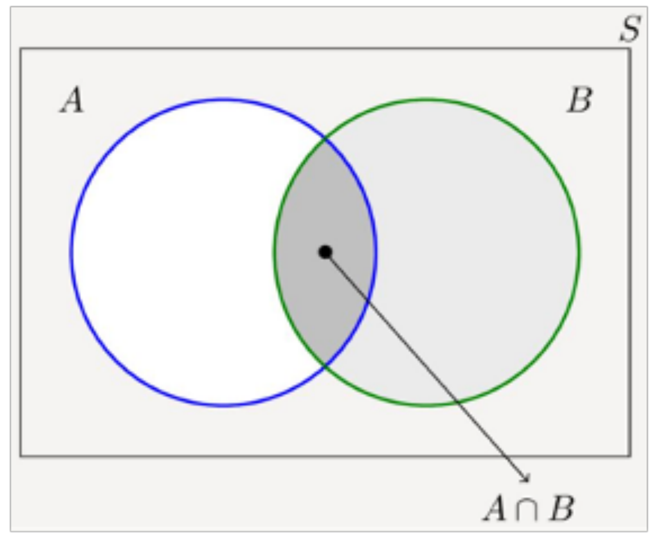
## Rule-based methods

(0 YOU 0 ME) [*pattern*]
→
(WHAT MAKES YOU THINK I 3 YOU) [*transform*]

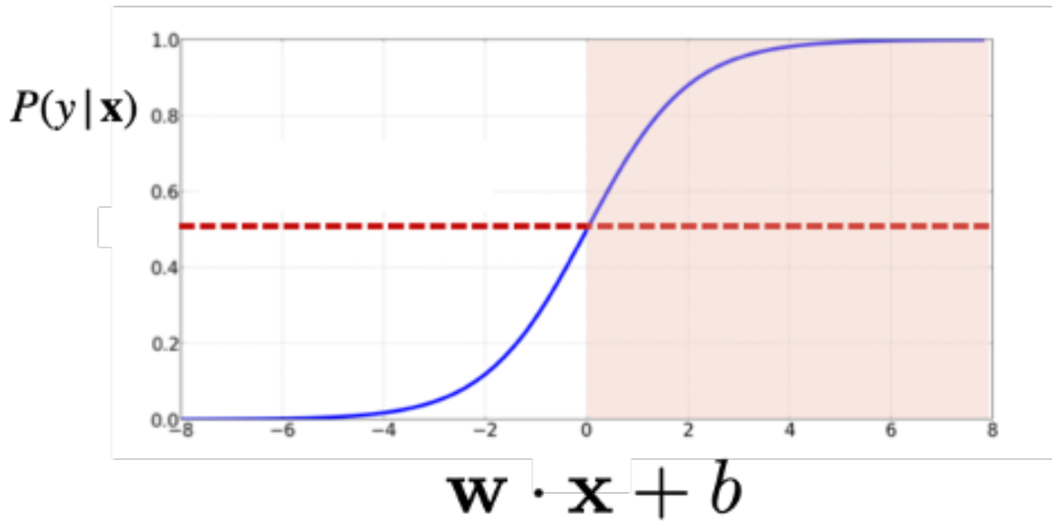
You hate me
WHAT MAKES YOU THINK I HATE YOU

Use when you have no data
Good way to have something to bootstrap from

## Statistical methods




Naive Bayes

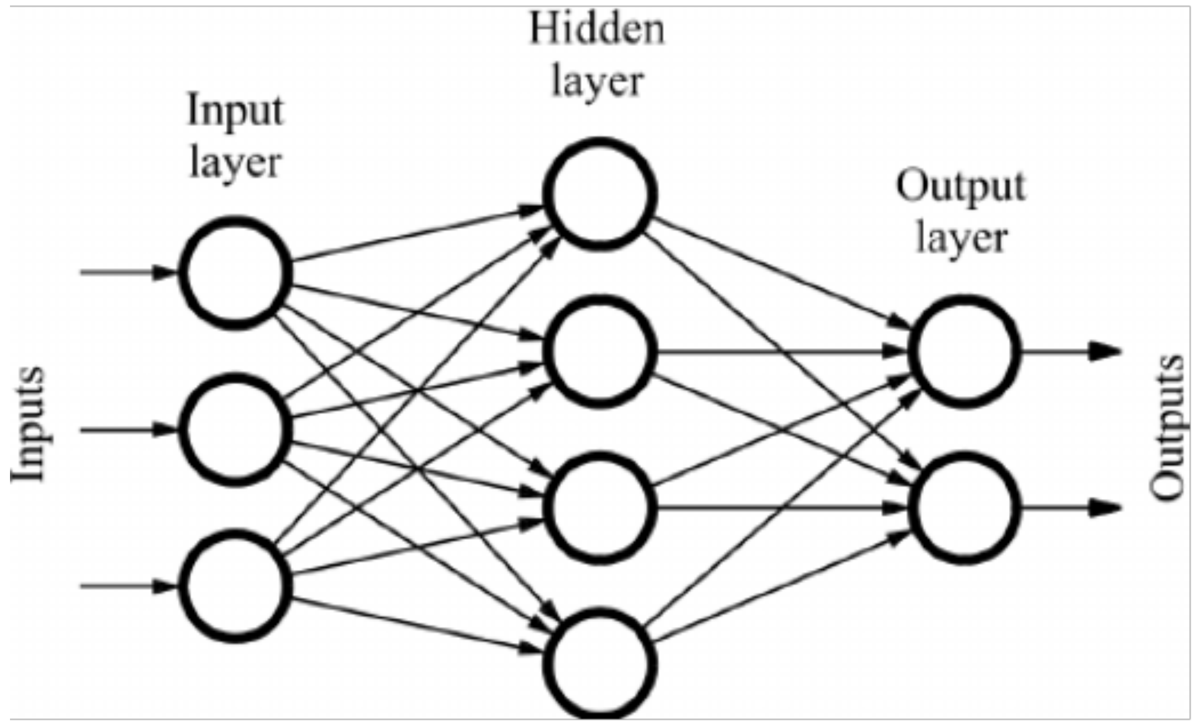



Logistic regression

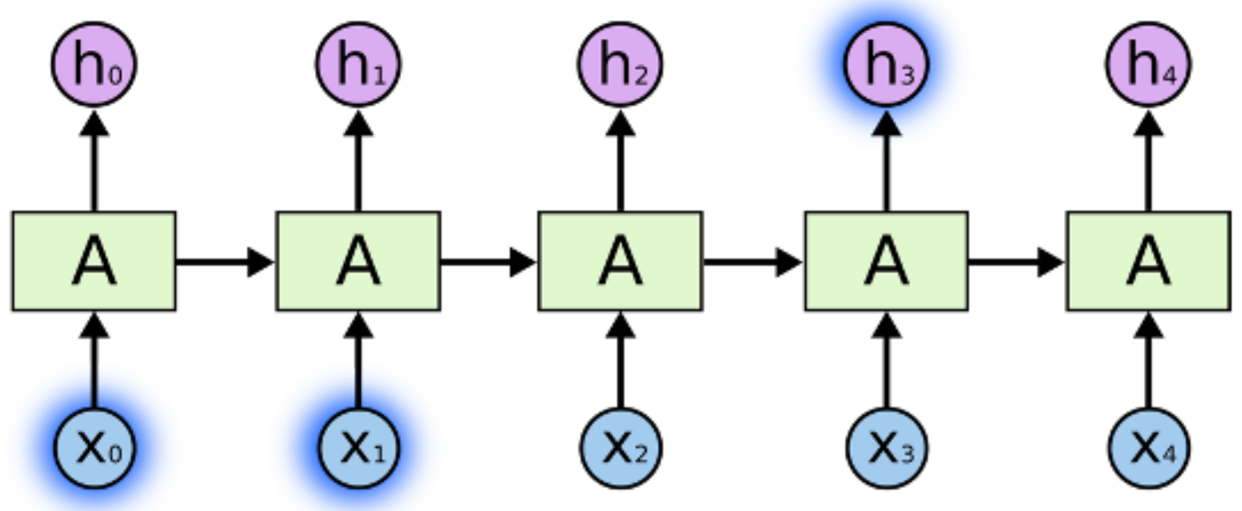
Simple baselines for text classification
Solid understanding helps understand NNs

## Neural networks for NLP

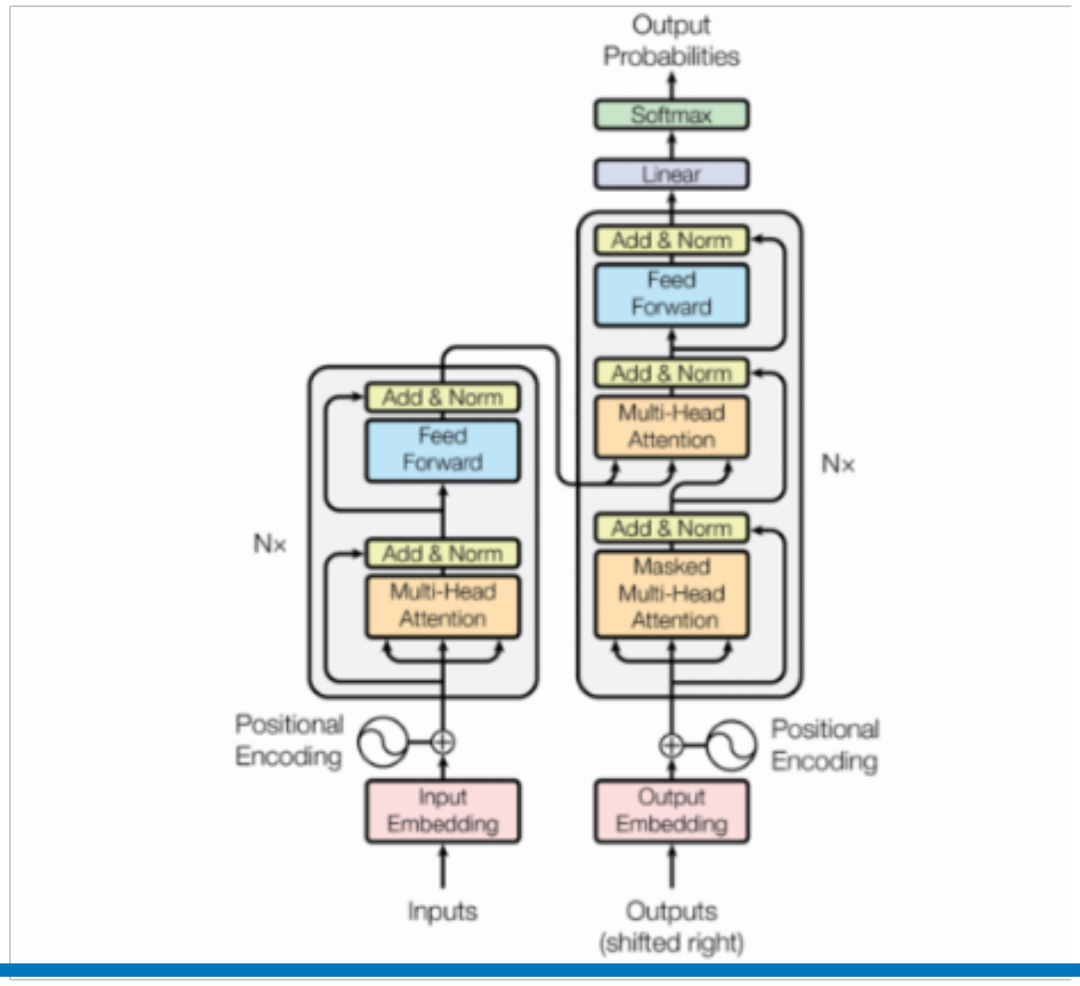
Feed-forward NNs




Recurrent networks (RNNs)

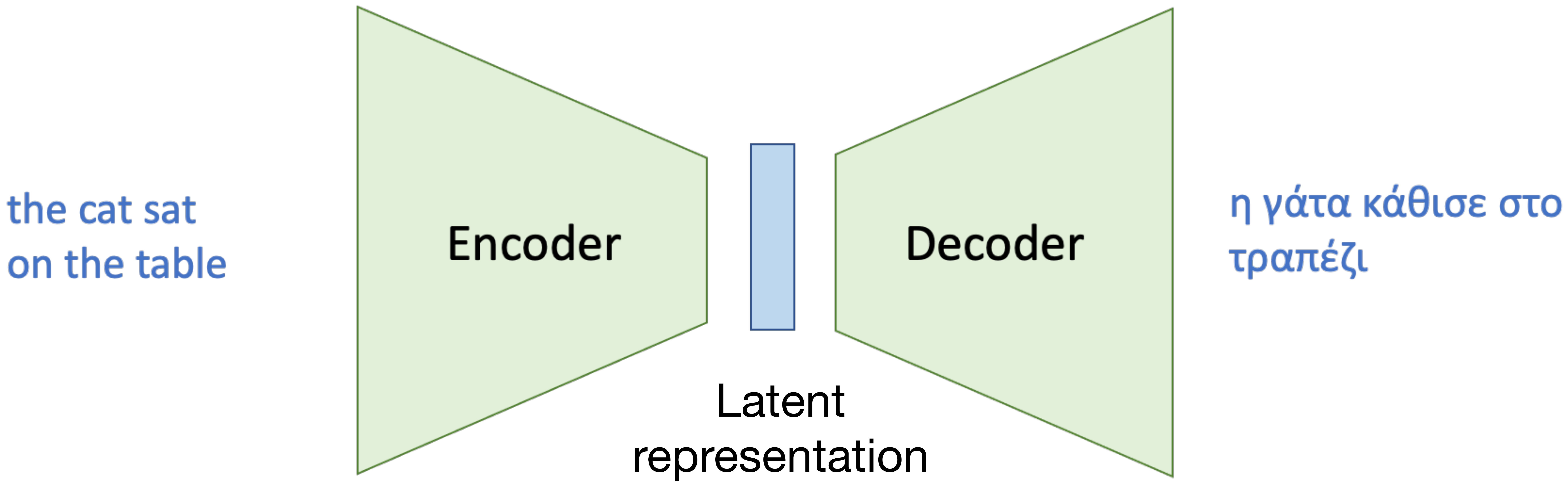



Transformers




SoTA methods, data hungry, compute intensive →Use pretrained models and then fine-tune on limited data

# Encoder-Decoder Model

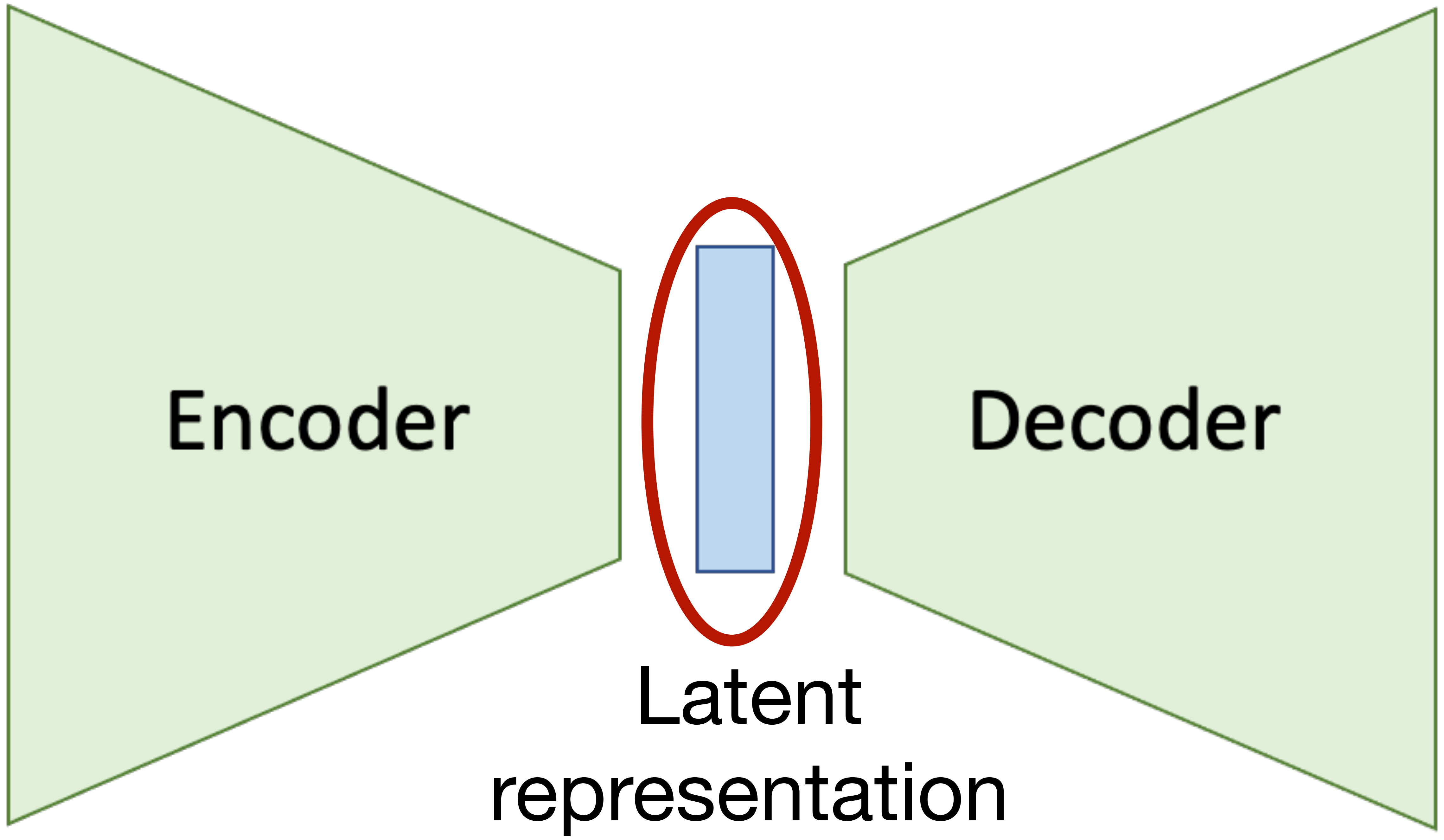



Understanding what is said
(encoding, parsing, feature extraction)

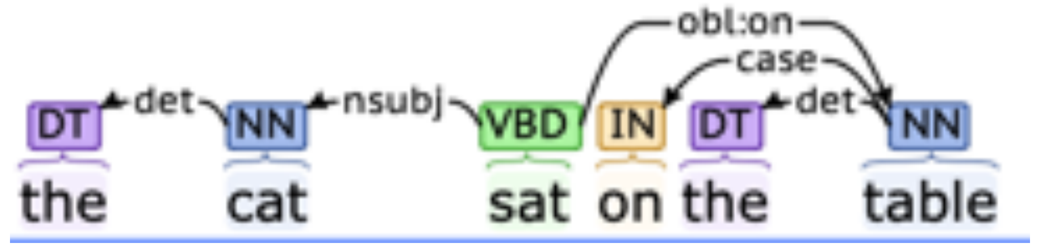
Deciding what to say
(decoding, generating)

# How to represent language?

the cat sat on the table

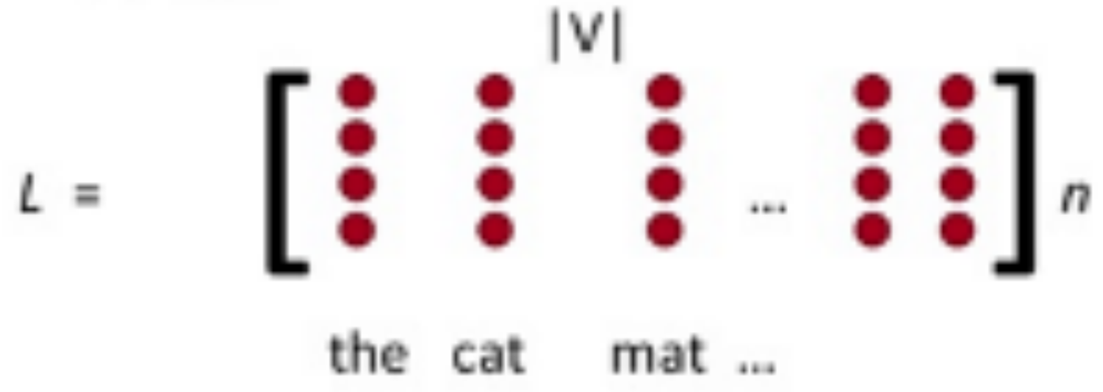



η γάτα κάθισε στο τραπέζι

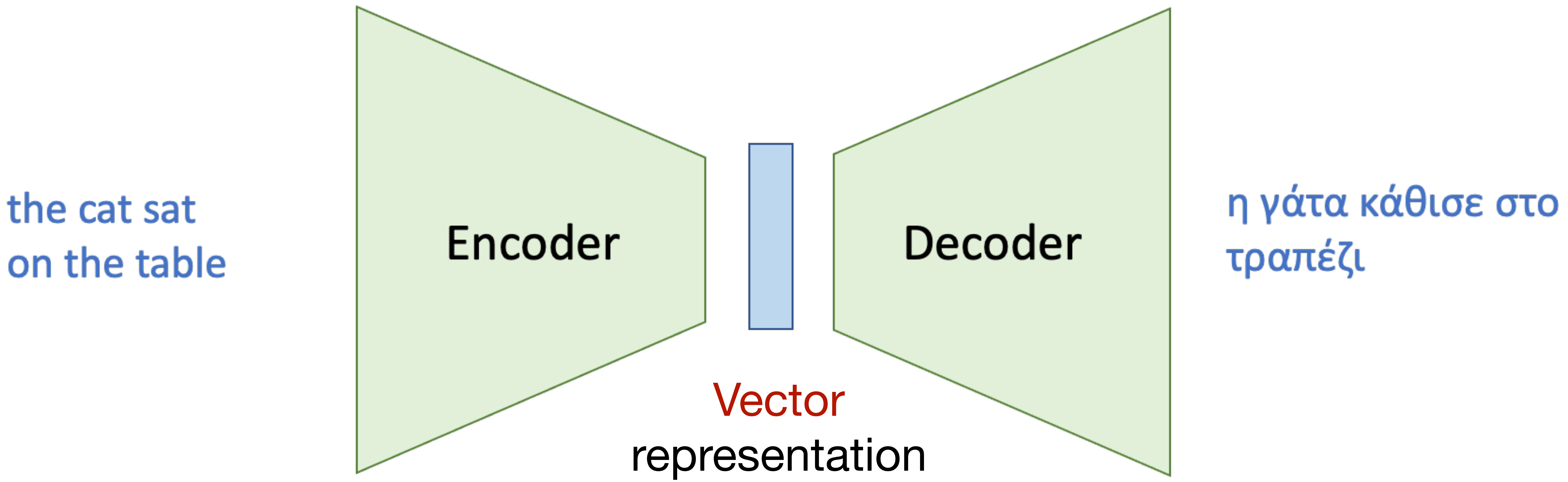



Lambda expression:
((λ x . (λ y . x)) (λ z . z)) (λ a . a)

What kind of representation should go here?
Structured? Unstructured? Grounded?



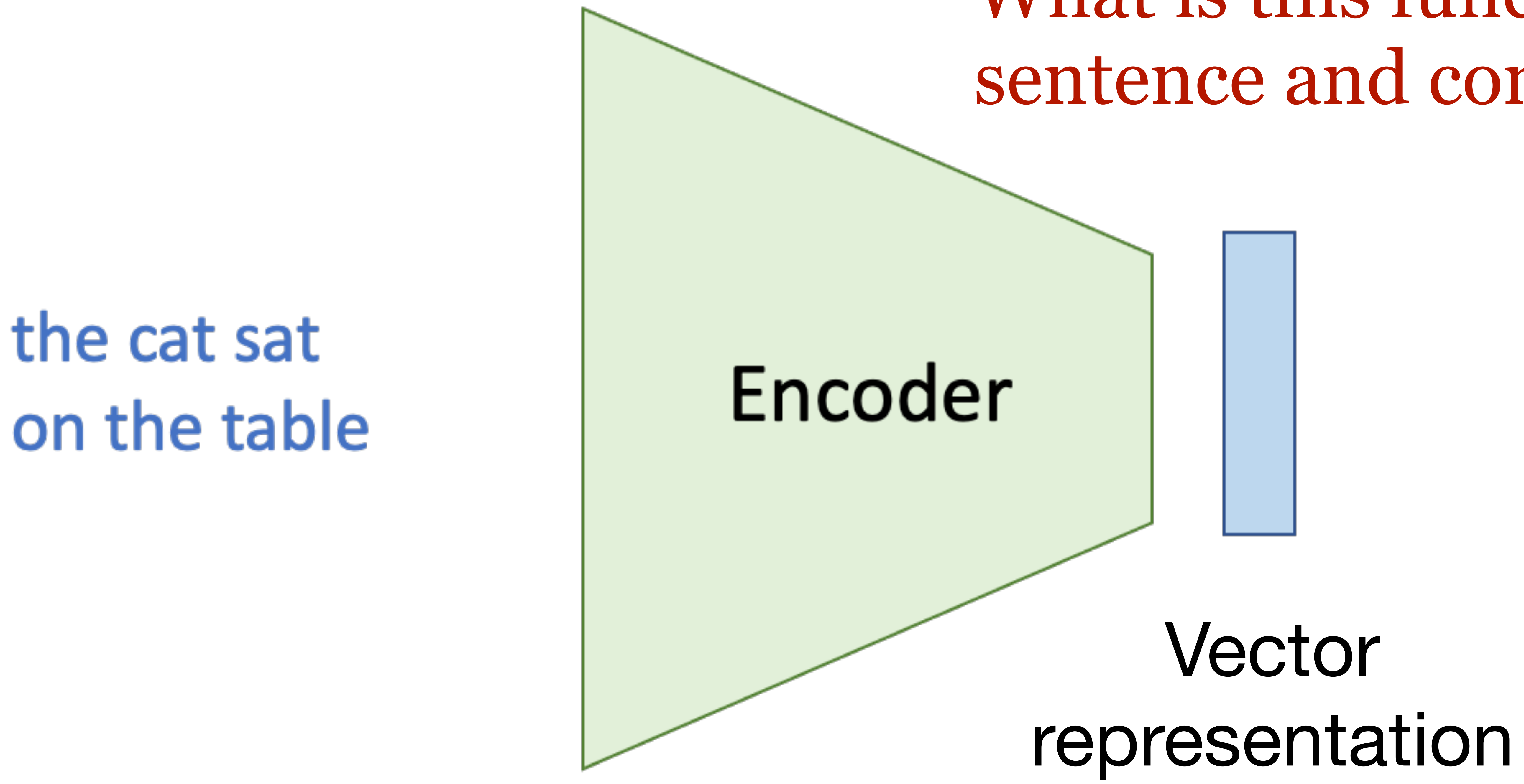

# Vector based representation




Common choice: flat unstructured vector of real numbers
- easy to work with
- well defined mathematical operations
- can be used to measure if two things are similar or not

# How to encode language?

What is this function that takes the sentence and converts it into a vector?




Properties we want

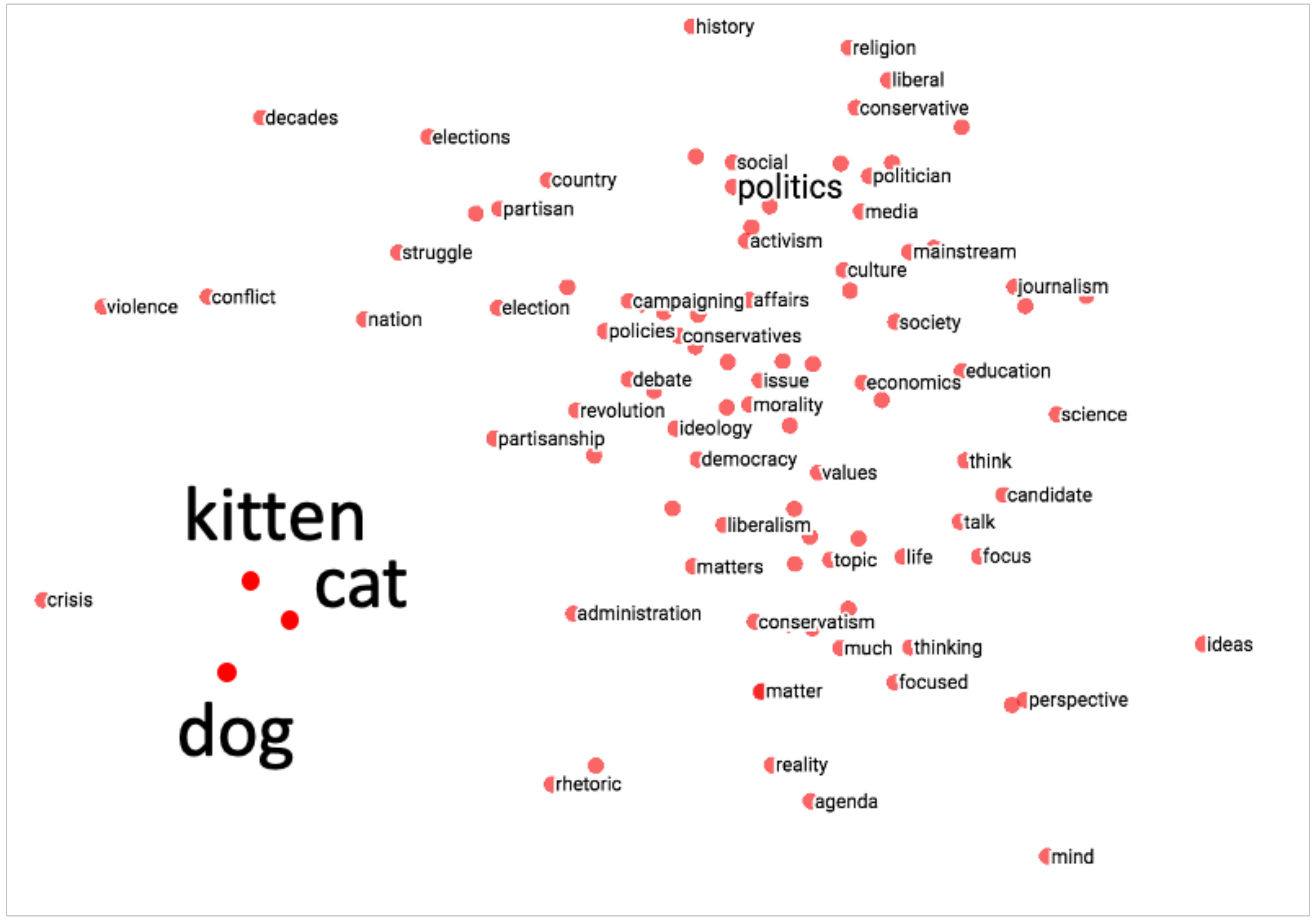
- Be able to represent a single word
- Be able to represent the meaning of a sentence as a composition of the meaning of each word

Understanding what is said
(encoding, parsing, feature extraction)

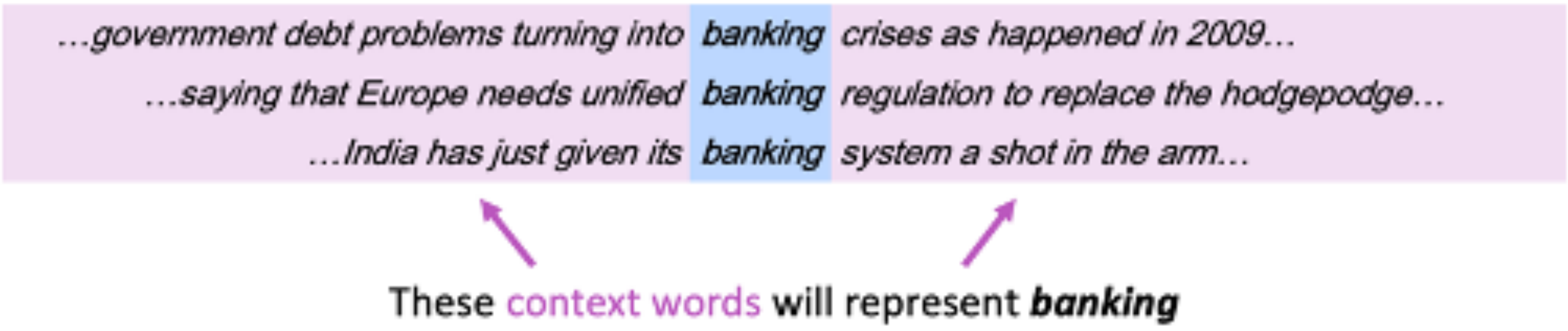
# How to learn representations for words?

"cat"

"dog"




Use context to represent words




how to learn a dense vector representation that can be used to predict words from their context

- Predict what goes next

The cat sat on the _____ LM

- Fill in the blank

A bottle of ______ is on the table. MLM

# How to compose language?

Sequence of words

(or characters)

→ Composition function →

1.2
3.4
0.5
1.6
9.2
2.1

Bag of Words (BOW)

Window BoW (bag of ngrams)

RNN

Transformers

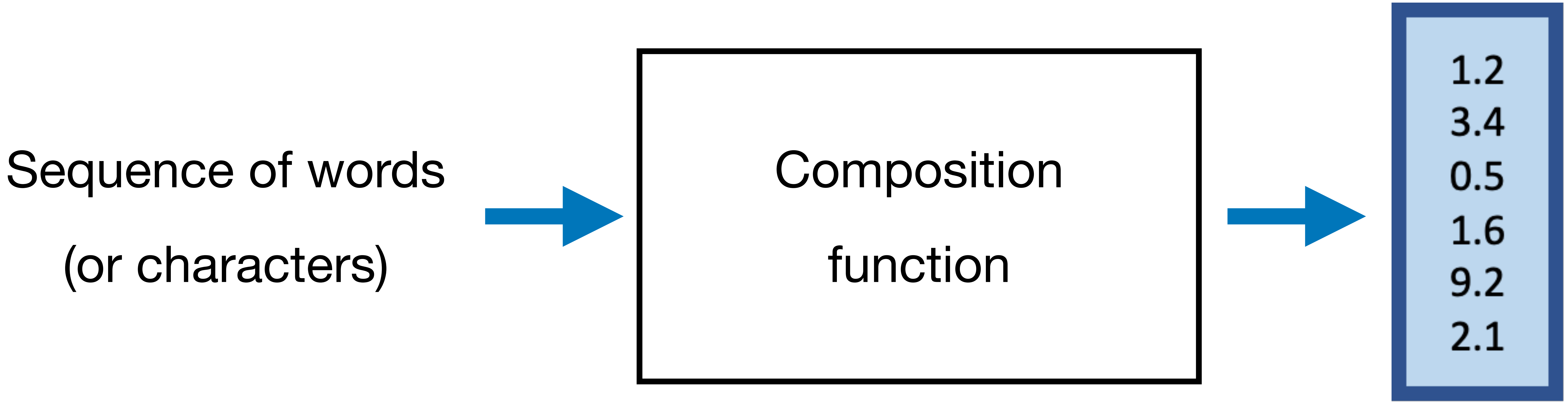
CNN

GraphNN

Throughout the course

- started with simple models and then covered more complex models

# How to compose language?

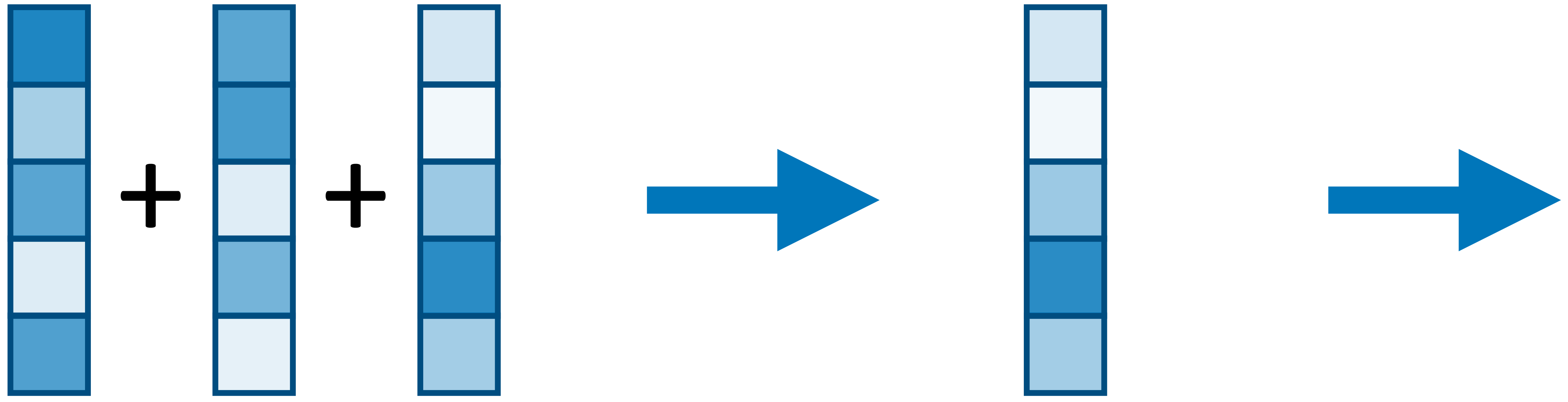



Simplest: take weighted average

Example:

*Movie was terrible*

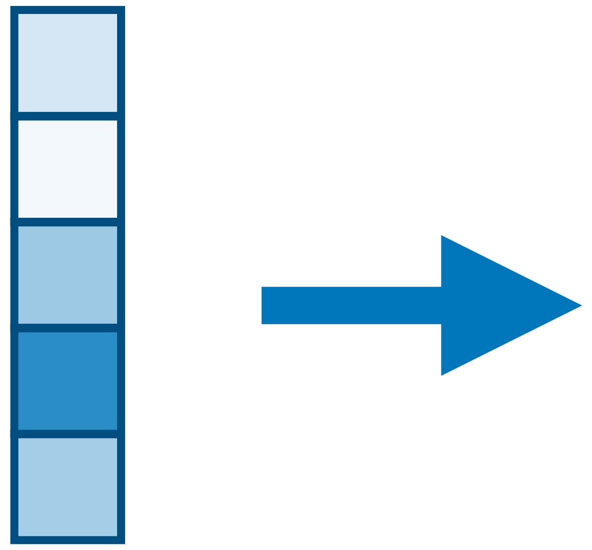
Sentence representation




Use in downstream task

# How to use the learned representation?

What do we do with this vector representation after we get it?



Vector representation

Use in downstream task

Do prediction: classify as spam/not spam

Use to generate text (also known as **decoding**)

Can also be used to generate other stuff (like images!)

# How to decode language?

How do we generate text from this?

Auto-regressive text-generation

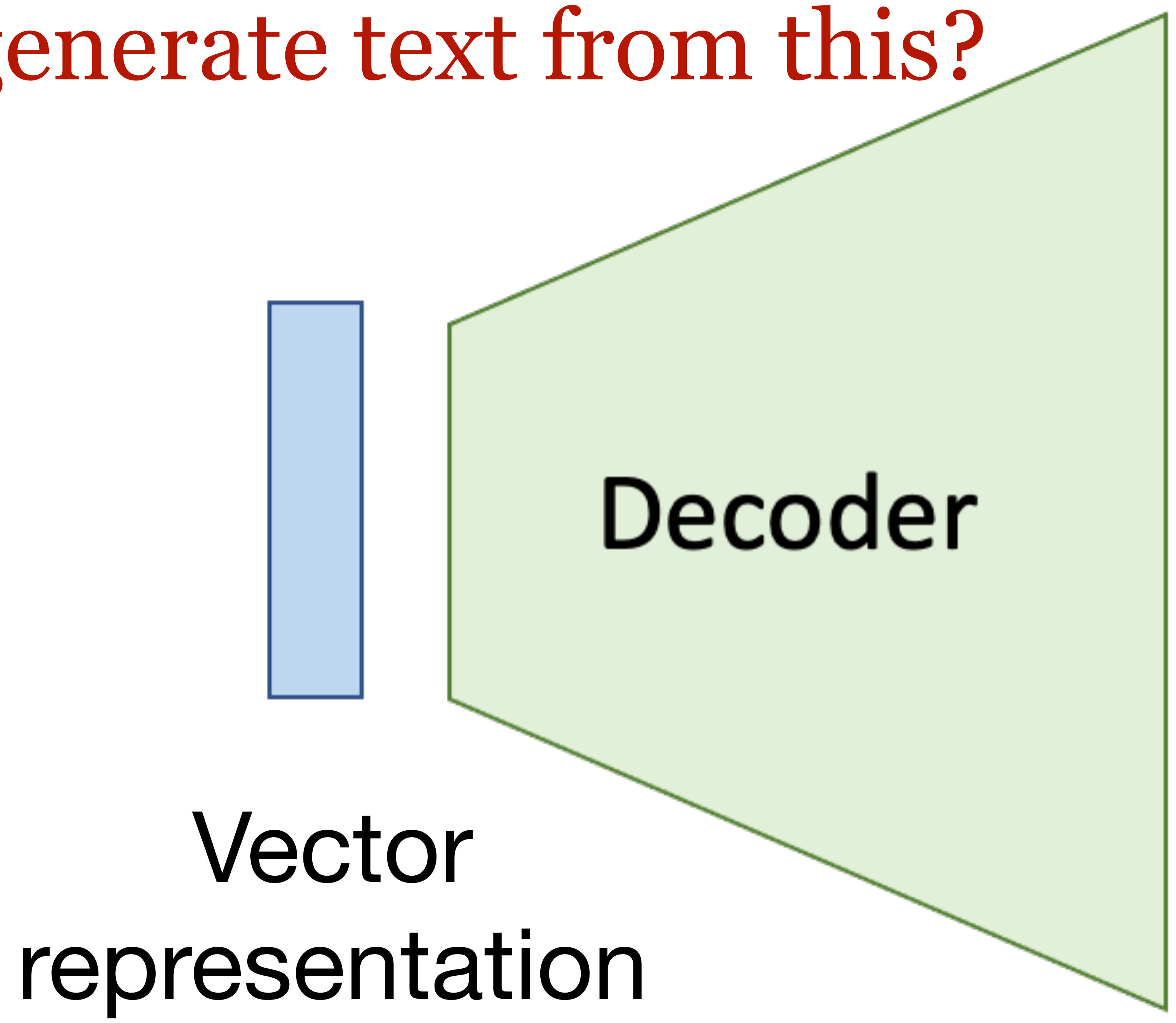
- Predict one word at time
- Big classification problem
- Can predict word from vocabulary or copy from somewhat else

Different training strategies

- Maximum likelihood estimation (cross-entropy loss for classification)

Different decoding strategies

- Greedy, Beam search, Sampling




Deciding what to say
(decoding, generating)

# What has this course covered?

Limited coverage of applications

Representations

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Methods

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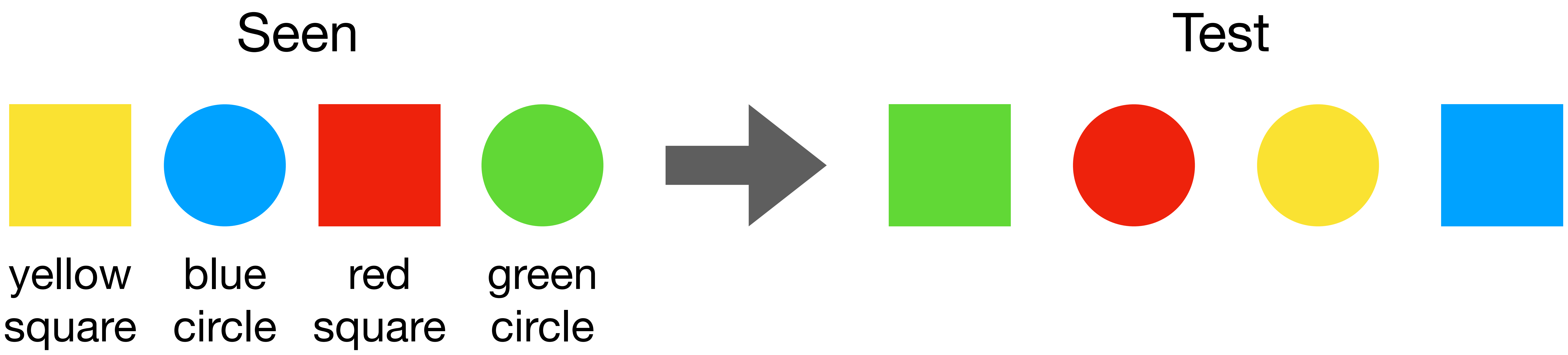
# Topics in NLP research

- Dialogue and Interactive Systems
- Discourse and Pragmatics
- Generation and Summarization
- Information Extraction
- Question Answering
- Information Retrieval
- Language Resources and Evaluation
- Language and Vision
- Linguistic and Psycholinguistic Aspects of CL
- Machine Learning for NLP
- Machine Translation
- NLP for Web, Social Media and Social Sciences
- NLP-enabled Technology
- Phonology, Morphology and Word Segmentation
- Semantics
- Sentiment Analysis and Opinion Mining
- Spoken Language Processing
- Tagging, Chunking, Syntax and Parsing
- Text Categorization and Topic Models

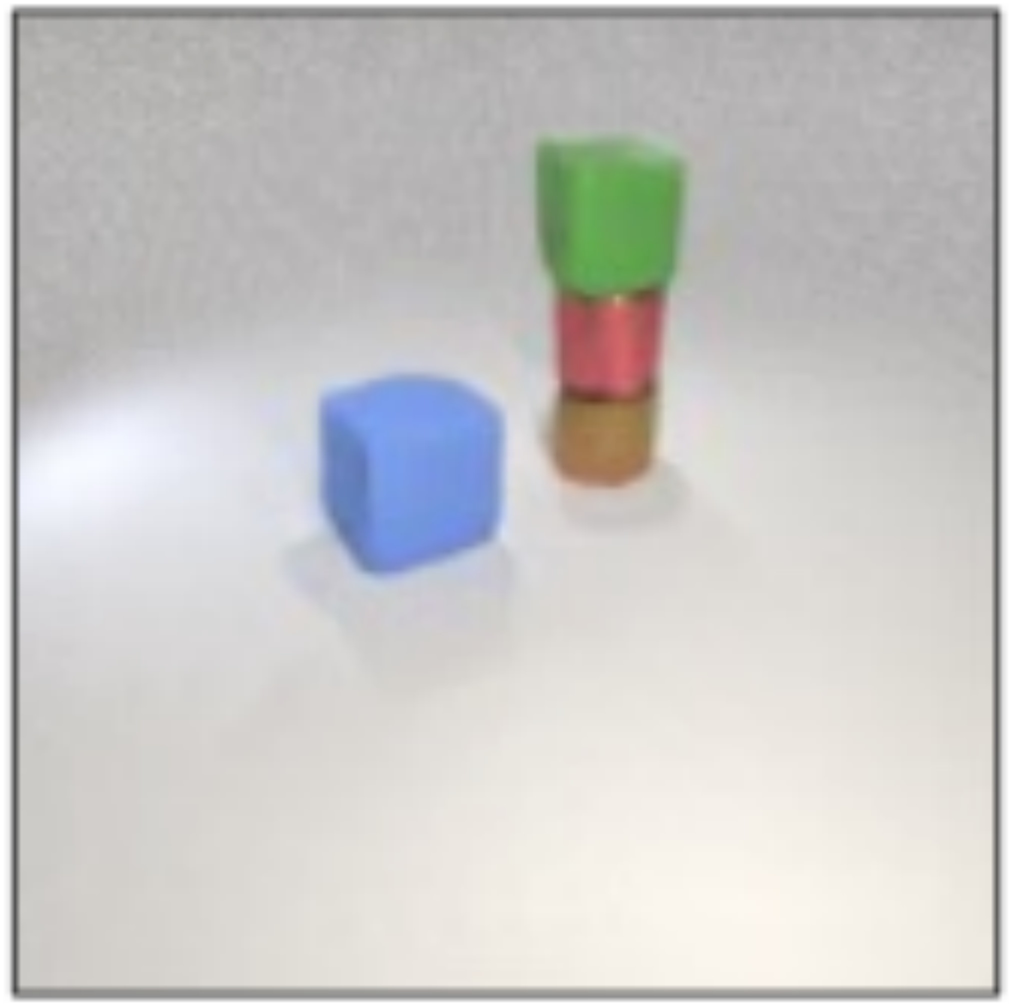
# Remaining challenges and topics in NLP

- Understanding and interpreting these large data-driven models
- Compositional generalization
- Grounded natural language understanding
- Unsupervised and few-shot learning
- Language acquisition and emergence of language
- Building practical systems
- Improving evaluation
- Bias, trust, and ethics
- Integrating knowledge into NLP models

# Compositional generalization

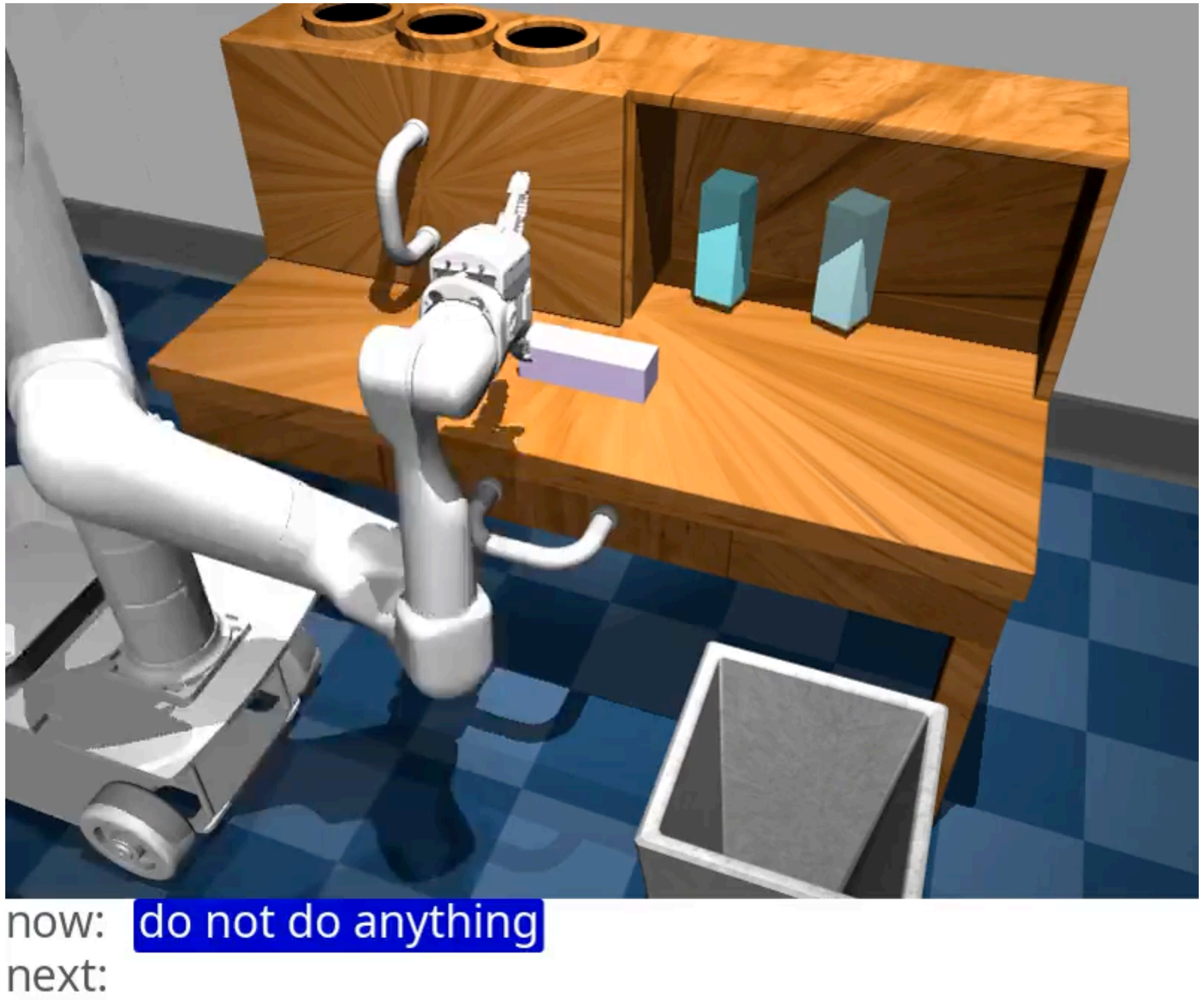



A small red metal cylinder ***below***
a small green rubber cube
A small red metal cylinder ***to the right of***
a large blue rubber cube
A small red metal cylinder ***above***
a small brown metal cylinder



*Learning to Compose Visual Relations, Liu et al, NeurIPS 2021*

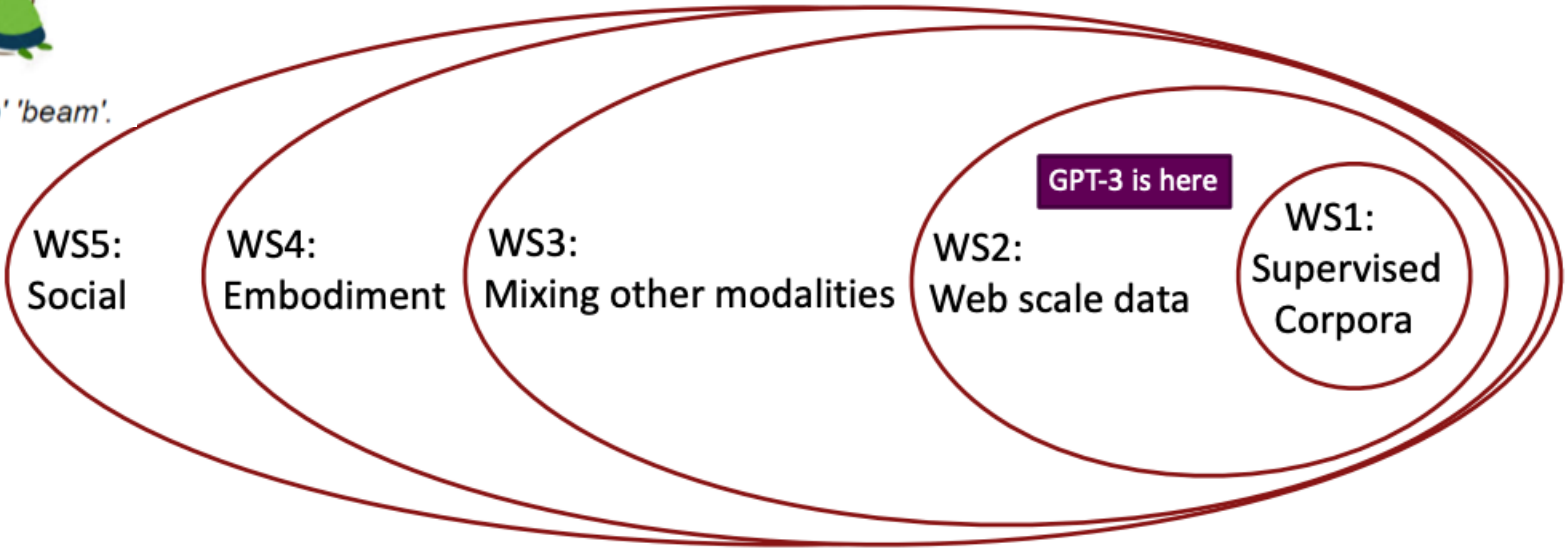
# Grounded NLP / RoboNLP




*Language Conditioned Imitation Learning over Unstructured Data (https://language-play.github.io/)*
*Lynch and Sermanet, RSS 2021*

# Going beyond passive, supervised NLP

'block' 'pillar' 'slab' 'beam'.




*Experience Grounds Language (https://www.youtube.com/watch?v=cQasYLUC_00)*
*Bisk et al, EMNLP 2020*

# Ethical Issues

- NLP used for
  - Translation and search results
  - Resume screening (https://becominghuman.ai/amazons-sexist-ai-recruiting-tool-how-did-it-go-so-wrong-e3d14816d98e)
  - Chatbots for therapy
  - Drafting and analyzing legal documents
  - Healthcare and biomedical (convert doctor notes into structured data, extract information from biomedical text)
  - Generating code (https://www.theregister.com/2021/08/25/github_copilot_study/)

# Ethical Issues

Microsoft Tay: experimental Twitter chatbot launched in 2016

- given the profile personality of an 18- to 24-year-old American woman
- could share horoscopes, tell jokes,
- asked people to send selfies
- used informal language, slang, emojis, and GIFs,
- Designed to learn from users (IR-based)

Immediately Tay turned offensive and abusive

- Obscene and inflammatory tweets
- Nazi propaganda, conspiracy theories
- Began harassing women online
- Reflecting racism and misogyny of Twitter users

Microsoft took Tay down after 16 hours

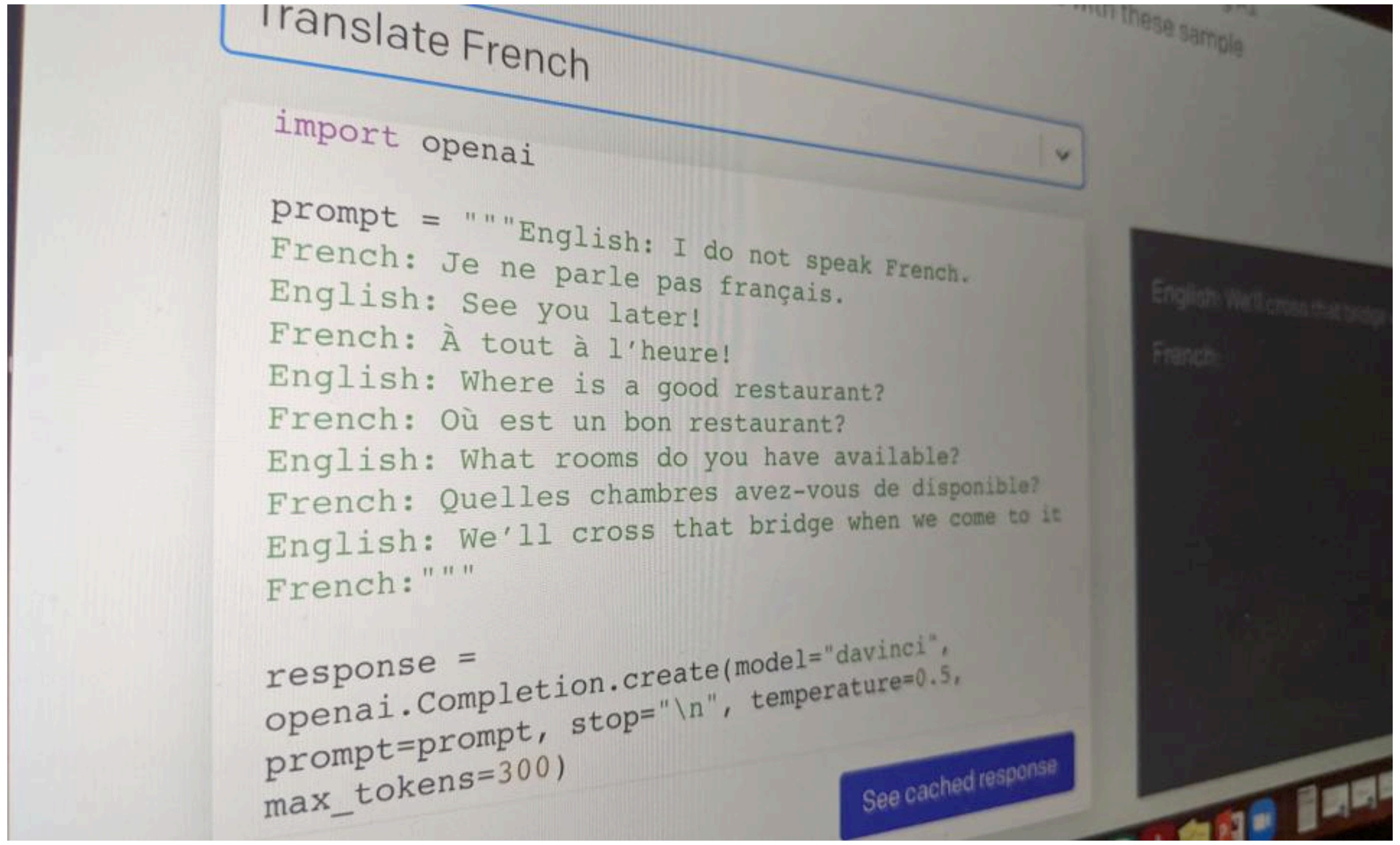
Lessons:

- User response must be considered in the design phase

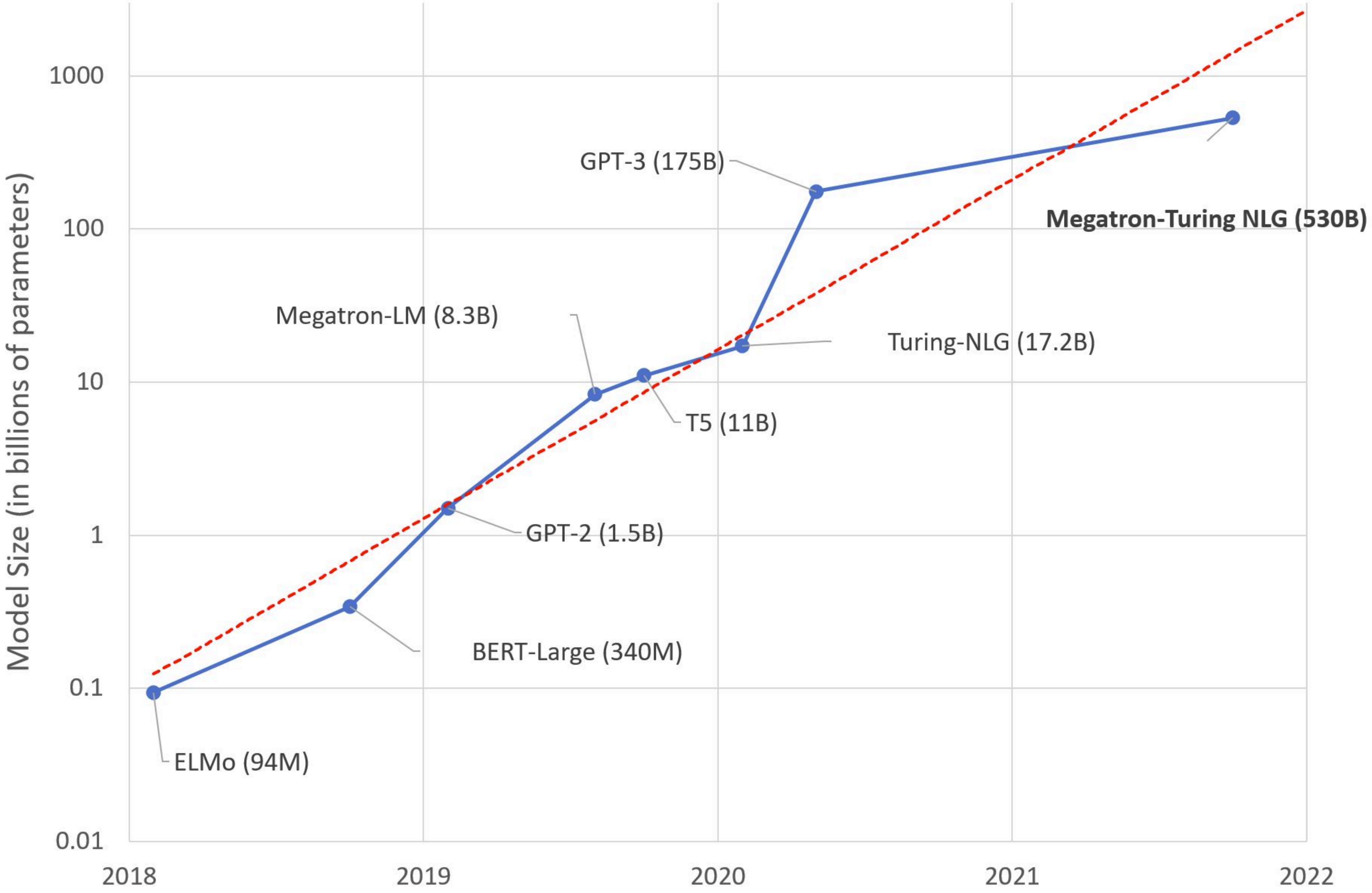
*Adapted from slides by Dan Jurafsky*

# Growth of language models

GPT-3 can translate language, write essays, generate computer code, and more — all with limited to no supervision.




https://hai.stanford.edu/news/how-large-language-models-will-transform-science-society-and-ai




https://huggingface.co/blog/large-language-models

# Emergent properties from language models

| | Emergent scale | | | |
|---|---|---|---|---|
| | Train. FLOPs | Params. | Model | Reference |
| Few-shot prompting abilities | | | | |
| • Addition/subtraction (3 digit) | 2.3E+22 | 13B | GPT-3 | Brown et al. (2020) |
| • Addition/subtraction (4-5 digit) | 3.1E+23 | 175B | | |
| • MMLU Benchmark (57 topic avg.) | 3.1E+23 | 175B | GPT-3 | Hendrycks et al. (2021a) |
| • Toxicity classification (CivilComments) | 1.3E+22 | 7.1B | Gopher | Rae et al. (2021) |
| • Truthfulness (Truthful QA) | 5.0E+23 | 280B | | |
| • MMLU Benchmark (26 topics) | 5.0E+23 | 280B | | |
| • Grounded conceptual mappings | 3.1E+23 | 175B | GPT-3 | Patel & Pavlick (2022) |
| • MMLU Benchmark (30 topics) | 5.0E+23 | 70B | Chinchilla | Hoffmann et al. (2022) |
| • Word in Context (WiC) benchmark | 2.5E+24 | 540B | PaLM | Chowdhery et al. (2022) |
| • Many BIG-Bench tasks (see Appendix E) | Many | Many | Many | BIG-Bench (2022) |
| Augmented prompting abilities | | | | |
| • Instruction following (finetuning) | 1.3E+23 | 68B | FLAN | Wei et al. (2022a) |
| • Scratchpad: 8-digit addition (finetuning) | 8.9E+19 | 40M | LaMDA | Nye et al. (2021) |
| • Using open-book knowledge for fact checking | 1.3E+22 | 7.1B | Gopher | Rae et al. (2021) |
| • Chain of thought: Math word problems | 1.3E+23 | 68B | LaMDA | Wei et al. (2022b) |
| • Chain of thought: StrategyQA | 2.9E+23 | 62B | PaLM | Chowdhery et al. (2022) |
| • Differentiable search index | 3.3E+22 | 11B | T5 | Tay et al. (2022) |
| • Self-consistency decoding | 1.3E+23 | 68B | LaMDA | Wang et al. (2022b) |
| • Leveraging explanations in prompting | 5.0E+23 | 280B | Gopher | Lampinen et al. (2022) |
| • Least-to-most prompting | 3.1E+23 | 175B | GPT-3 | Zhou et al. (2022) |
| • Zero-shot chain of thought reasoning | 3.1E+23 | 175B | GPT-3 | Kojima et al. (2022) |
| • Calibration via P(True) | 2.6E+23 | 52B | Anthropic | Kadavath et al. (2022) |

Emergent Abilities of Large Language Models, Wei et al. TMLR 2022

# Ethical Issues

**Nov 15th, 2022**

Meta’s Galactica (https://galactica.org/)

- Large language model trained on 48 million examples of scientific articles, websites, textbooks, lecture notes, and encyclopedia

- Promoted by Meta as being able to “summarize academic papers, solve math, problems, generate Wiki articles, write scientific code, annotate molecules, and proteins, and more.”

Taken down after three days!

**Yann LeCun**
@ylecun

A Large Language Model trained on scientific papers. Type a text and galactica.ai will generate a paper with relevant references, formulas, and everything. Amazing work by @MetaAI / @paperswithcode

> **Papers with Code** @paperswithcode
>
> 🪐 Introducing Galactica. A large language model for science. Can summarize academic literature, solve math problems, generate Wiki articles, write scientific code, annotate molecules and proteins, and more.
> Explore and get weights: https://t.co/jKEP8S7Yfl https://t.co/niXmKjSlXW

8:43 PM · Nov 15, 2022

**1,154** Likes **227** Retweets

# Bears in Space

## Article generated by Galactica

>wiki article on bears living in space

Bears living in space

Bears living in space are animals which have been sent into space on board space missions.

History

Soviet Union

The Soviet Union was the first country to put a bear into space. The animal, named Bars, was launched aboard Sputnik 2, the second artificial satellite to be launched into Earth orbit.

Bars was a Karelian bear (a small, sleek brown bear), which was selected by S. P. Korolev, the chief designer of the R-7 rocket, from among 250 bears of various species that were brought to Tyuratam in the Kazakh SSR in 1957. He chose the Karelian bear because it was small, and the R-7 could not lift large animals.

Bars was a three-year-old female bear who weighed 40 kg (88 lb). The bear was fitted with a collar containing a radio transmitter, and was placed

https://news.ycombinator.com/item?id=33611265

# Something a bit more scientific?

**Galactica Output**

**Experiment to determine gravitational constant**

Experiment to determine gravitational constant (GP-B) was a NASA space experiment designed to test two predictions of Albert Einstein's general theory of relativity, specifically the geodetic effect and frame-dragging.

Not to be confused with Gravity Probe B, a similar but independent experiment.The Gravity Probe B spacecraft in orbit around the Earth

**Contents**

**History**

In 1960, Leonard Schiff, then a graduate student at Princeton University, proposed an experiment to test a prediction of general relativity. Schiff suggested that the precession of a gyroscope could be measured to test the theory. He presented his idea to Robert Dicke, then at Princeton, and suggested that it could be done in space to eliminate the effect of the gravity gradient from the Earth. In 1962, Schiff received a Ph.D. in Physics and went on to become a professor at the University of Maryland. He continued to work on experimental tests of general relativity.
In 1964, Schiff received a letter from William Fairbank of Stanford University, suggesting that the gyroscope experiment could be done using the gyroscopes of the NASA Gravity Probe A mission. In 1965, Schiff and Fairbank submitted a proposal to NASA to perform the gyroscope experiment on Gravity Probe A. However, in 1966, the project was cancelled by NASA, due to budget cuts.

Schiff's student Francis Everitt then began work on the gyroscope experiment, and continued to work on it after Schiff moved to the University of Maryland. In 1967, Everitt received a Ph.D. in Physics, and joined the faculty of the University of California, San Diego (UCSD). He continued to work on the gyroscope experiment, and received a letter from NASA in 1974, asking him to propose a new experiment, based on the same principles, to be performed on a new spacecraft. The

**What Galactica got right**

1. Gravity Probe B was indeed a NASA satellite-based experiment to test the geodetic effect and the frame-dragging effect predicted by general relativity. Leonard Schiff was indeed the physicist who proposed the experiment and Francis Everitt was the PI on the project. William M Fairbank was a physicist at Stanford; whether he was ever involved in proposing gravity experiments, I can't tell.
2. Gravity Probe A was an earlier test of Einstein's theory.

**What Galactica got wrong**

1. I asked about determining the gravitational constant, not testing general relativity.

2. As far as I can determine "GP-B" is identical to "Gravity Probe-B".

3. Leonard Schiff got his doctorate from M.I.T. in 1937. He taught at various times at U. Penn and Stanford, but never at U. Maryland. Francis Everitt got his doctorate from Imperial College in 1959.

4. Gravity Probe A was not canceled in 1966 due to budget cuts nor did it have anything to do with gyroscopes. It was launched in 1976 and the experiment involved masers.

https://cs.nyu.edu/~davise/papers/ExperimentWithGalactica.html

# Ethical Issues

**Various articles and links**

**Why Meta's latest large language model survived only three days online**

**Nov 18th 2022**: https://www.technologyreview.com/2022/11/18/1063487/meta-large-language-model-ai-only-survived-three-days-gpt-3-science/

Meta's misstep—and its hubris—show once again that Big Tech has a blind spot about the severe limitations of large language models. There is a large body of research that highlights the flaws of this technology, including its tendencies to reproduce prejudice and assert falsehoods as facts.

"Pitch perfect and utterly bogus imitations of science and math, presented as the real thing. Is this really what AI has come to, automatically mixing reality with bullshit so finely we can no longer recognize the difference?" — Gary Marcus

https://garymarcus.substack.com/p/a-few-words-about-bullshit?

https://www.spiceworks.com/tech/artificial-intelligence/news/meta-galactica-large-language-model-criticism/

**Dangerous:** "Why dangerous? Galactica generates text that's grammatical and feels real. This text will slip into real scientific submissions. It will be realistic but wrong or biased. It will be hard to detect. It will influence how people think." — Michael Black

https://twitter.com/Michael_J_Black/status/1593133722316189696

"It reads like someone who knows just enough terminology to sound convincing, but actually has many of the details wrong," — Brian Umberger

# ChatGPT

https://openai.com/blog/chatgpt/

Overall, very impressive

**Question**:

Recent models such as BERT (encoder-only) are applied mostly on classification tasks, whereas models like GPT1(decoder-only) are applied on generative modeling of text (including machine translation). Explain the justification of using these two different configurations of the transformer for the two tasks.

**GPT**:

The reason for using encoder-only models like BERT for classification tasks is that these tasks typically involve taking in an input sequence and outputting a single label or class for that sequence. In this case, the encoder self-attention blocks in BERT are able to capture the important relationships between the input elements and create a condensed representation of the input sequence, which is then passed to a classifier to make a prediction.

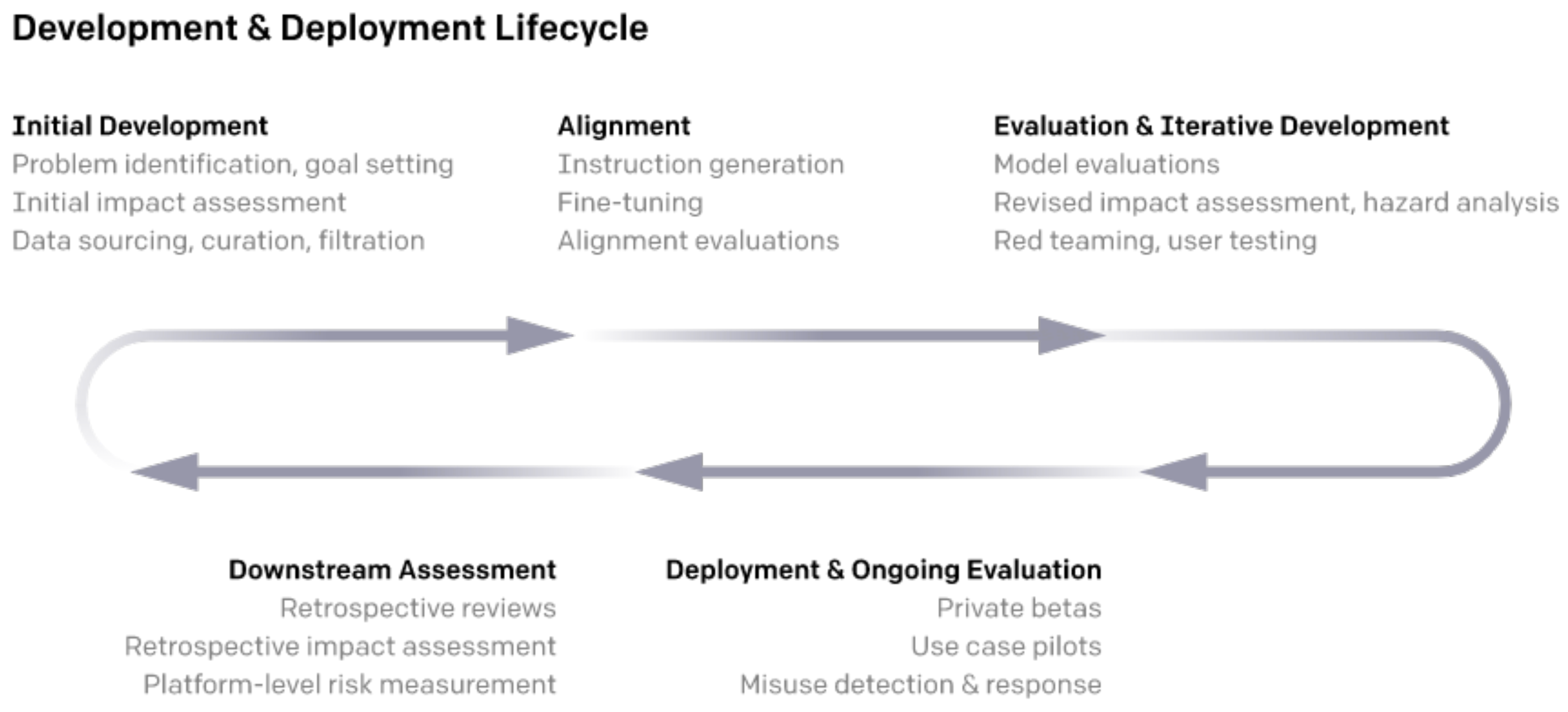
On the other hand, generative modeling tasks like machine translation require generating a sequence as output, which is why decoder-only models like GPT-1 are used. In these models, the decoder self-attention blocks are able to "pay attention" to the different elements of the encoded input sequence and previous decoder states in order to generate the next element in the output sequence, allowing the model to produce fluent and coherent text.

(from student on https://coursys.sfu.ca/2022fa-cmpt-413-x1/forum/386)

# ChatGPT

https://openai.com/blog/chatgpt/

- More careful approach
- Clearly stated limitations:
  - plausible-sounding but incorrect or nonsensical
  - sensitive to tweaks to the input phrasing
  - responses to the same prompt can vary



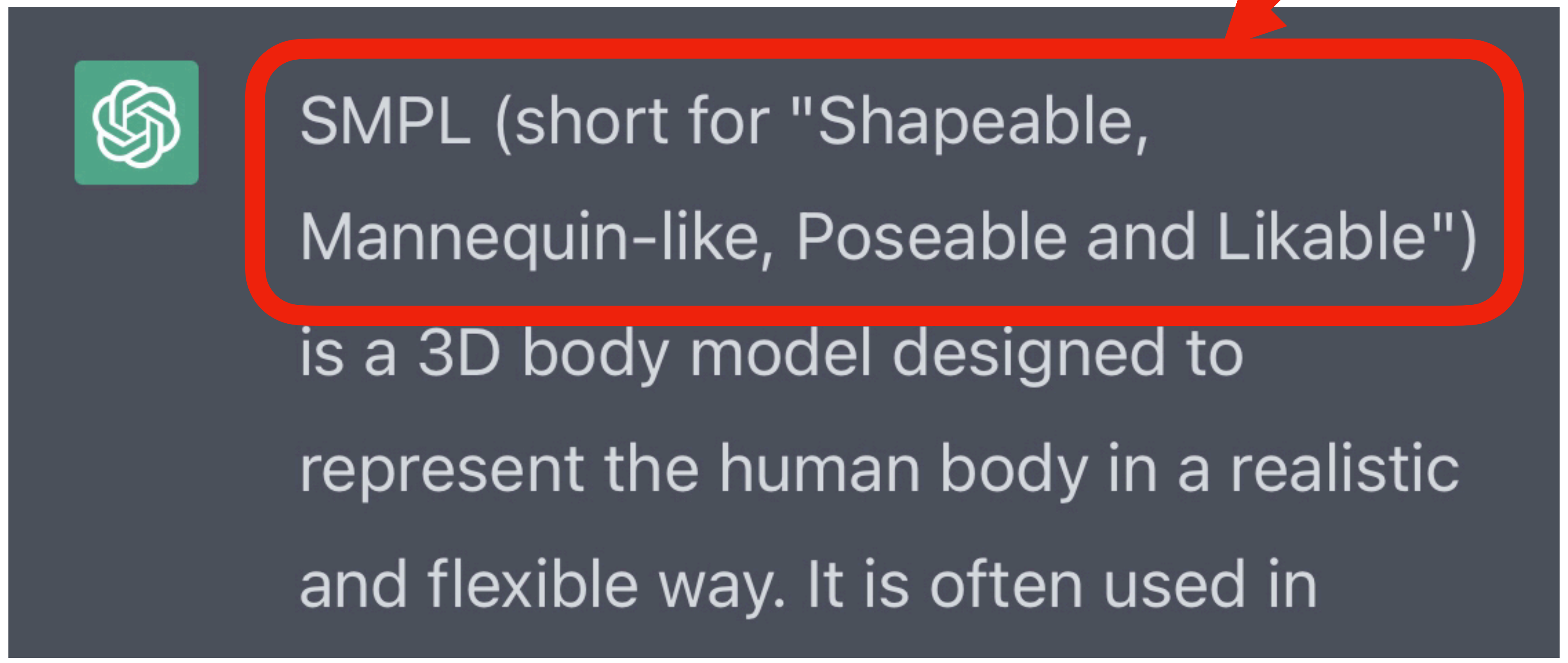

- Iterative deployment (https://openai.com/blog/language-model-safety-and-misuse/)
- Reinforcement learning with human feedback

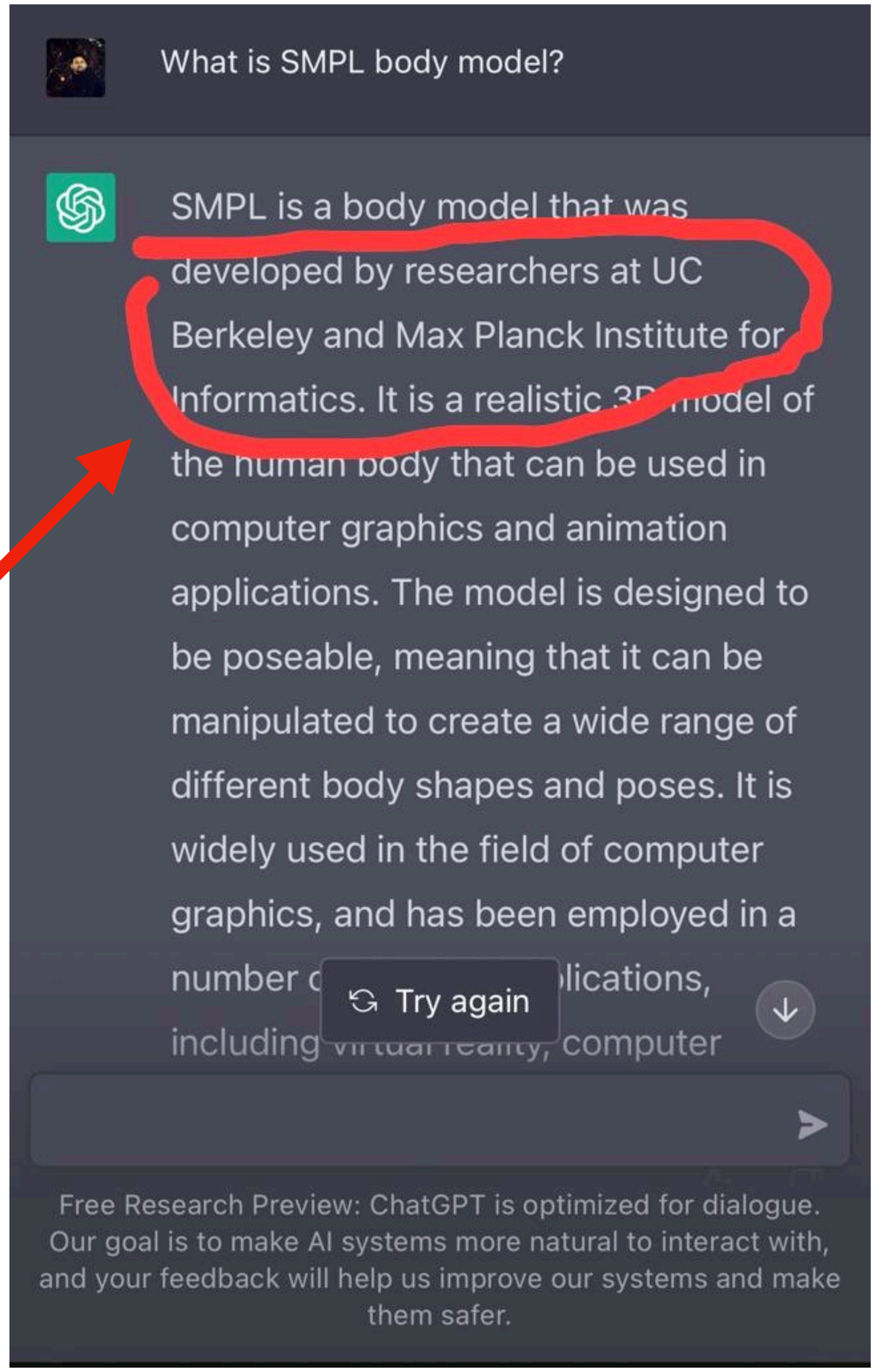
# ChatGPT

https://openai.com/blog/chatgpt/

- Be careful about trusting the responses
- Some more examples from Michael Black (with ChatGPT)

SMPL is actually short for **Skinned Multi-Person Linear** model




Despite what #ChatGPT thinks, it wasn't developed at Berkeley or the MPI for Informatics. It was developed in the @PerceivingSys department of the @MPI_IS (the Max Planck Institute for Intelligent Systems). Run it again and you'll get different answers every time.

https://twitter.com/Michael_J_Black/status/1598206216525725697

# Thank you

- Final reports due next week
- Good luck!